STAT 6820: Assignment 6

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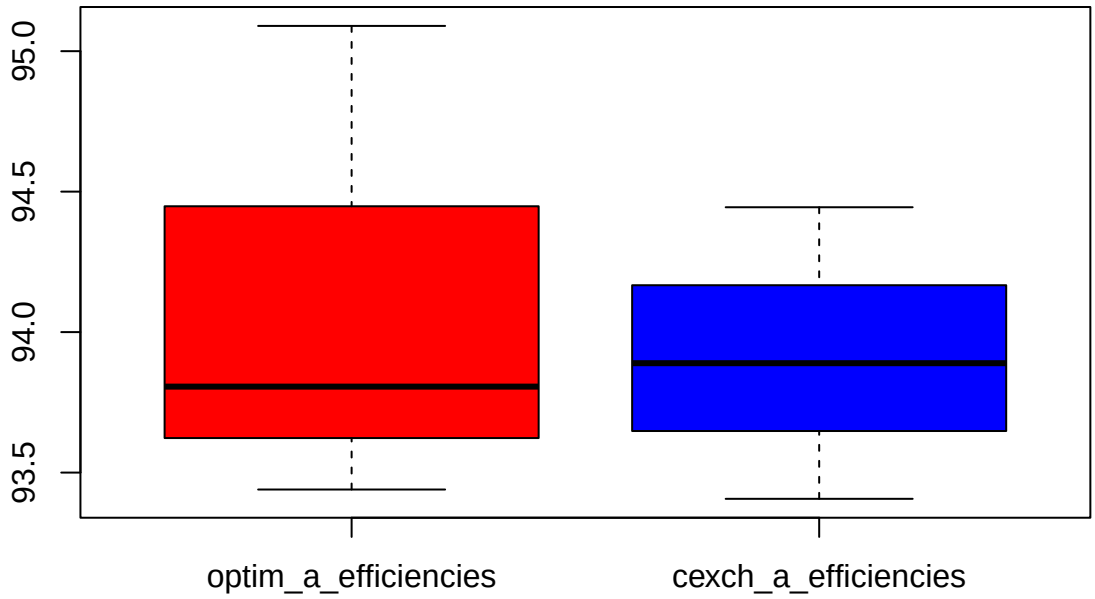
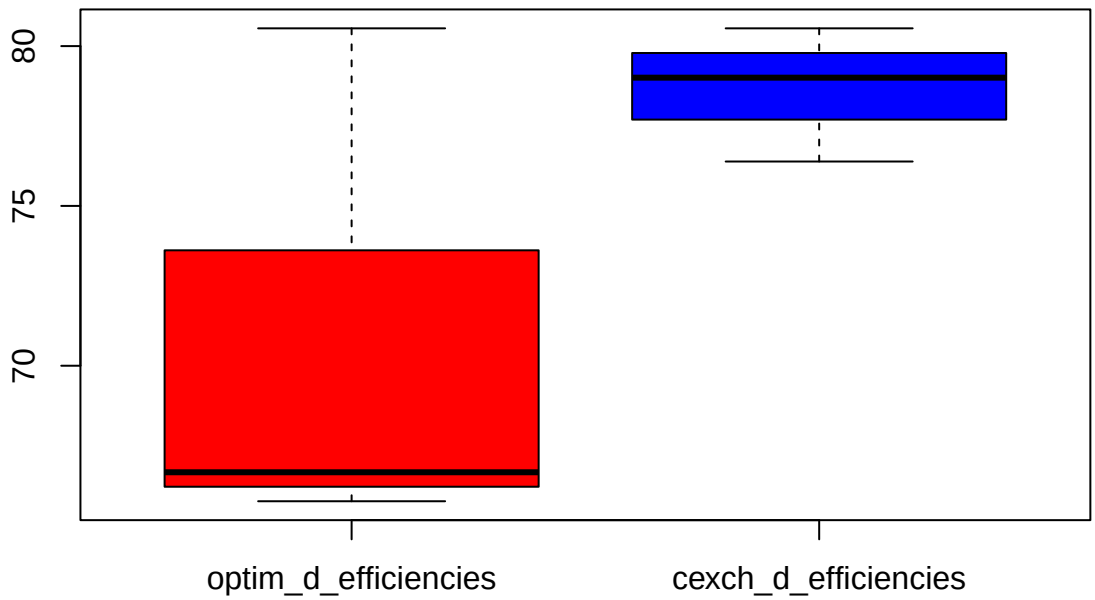
2023-03-30

Problem 1:

3 trials were run computing the A and D score using the built-in optimization routine *optim* as well as my implementation of coordinate exchange. A boxplot is given below visualizing the efficiency of the two algorithms, using Peter & Goos proposed design as a baseline for comparison.





** WRITE ANALYSIS **

Problem 2:

Here we derive the region moments matrices for the $K = 1, 2$ settings.

For the $K = 1$ setting, we compute

$$\mathbf{M} = \int_{\mathbf{x}\in[-1,+1]^2} \mathbf{f}(\mathbf{x})\mathbf{f}'(\mathbf{x})d\mathbf{x}$$

$$= \int_{-1}^{+1} \begin{bmatrix} 1 & x & x^2 \\ x & x^2 & x^3 \\ x^2 & x^3 & x^4 \end{bmatrix} dx$$

$$= \begin{bmatrix} x & \frac{x^2}{2} & \frac{x^3}{3} \\ \frac{x^2}{2} & \frac{x^3}{3} & \frac{x^4}{4} \\ \frac{x^3}{3} & \frac{x^4}{4} & \frac{x^5}{5} \end{bmatrix} \Big|_{-1}^{+1}$$

$$= \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix} - \begin{bmatrix} -1 & \frac{1}{2} & -\frac{1}{3} \\ \frac{1}{2} & -\frac{1}{3} & \frac{1}{4} \\ -\frac{1}{3} & \frac{1}{4} & -\frac{1}{5} \end{bmatrix} = \begin{bmatrix} 2 & 0 & \frac{2}{3} \\ 0 & \frac{2}{3} & 0 \\ \frac{2}{3} & 0 & \frac{2}{5} \end{bmatrix}$$

For the $K = 2$ setting, we compute

$$\mathbf{M} = \int_{\mathbf{x}\in[-1,+1]^2} \mathbf{f}(\mathbf{x})\mathbf{f}'(\mathbf{x})d\mathbf{x}$$

$$= \int_{-1}^{+1}\int_{-1}^{+1} \begin{bmatrix} 1 & x_1 & x_2 & x_1x_2 & x_1^2 & x_2^2 \\ x_1 & x_1^2 & x_1x_2 & x_1^2x_2 & x_1^3 & x_1x_2^2 \\ x_2 & x_1x_2 & x_2^2 & x_1x_2^2 & x_1^2x_2 & x_2^3 \\ x_1x_2 & x_1^2x_2 & x_1x_2^2 & x_1^2x_2^2 & x_1^3x_2 & x_1x_2^3 \\ x_1^2 & x_1^3 & x_1^2x_2 & x_1^3x_2 & x_1^4 & x_1^2x_2^2 \\ x_2^2 & x_1x_2^2 & x_2^3 & x_1x_2^3 & x_1^2x_2^2 & x_2^4 \end{bmatrix} dx_1dx_2$$

Problem 3:

For the $K = 1$, $N = 5$ setting, coordinate exchange was run 50 times. Despite a large number of attempts, there were only 22 *unique* proposed designs. There were even fewer unique i-scores, just 6. In fact, 44% (22 total) of the proposed designs achieved the minimum . The prevalence of equivalent designs makes recommendation challenging, but a design that achieved the minimum i-score (0.8885422) was

$$\mathbf{X} = \begin{bmatrix} -1.0 \\ 1.0 \\ -0.1 \\ -0.1 \\ 0.2 \end{bmatrix}$$

For the $K = 2$, $N = 10$ setting we perform the same computations. Similar results were obtained. There were only 22 *unique* proposed designs and just 19 unique i values.9% (9 total) of the proposed designs achieved the minimum . A design that achieved the minimum i-score (1.9059201) was

$$\mathbf{X} = \begin{bmatrix} 0.1 & 0.1 \\ 1.0 & -1.0 \\ -0.3 & -1.0 \\ -1.0 & 1.0 \\ -0.3 & -1.0 \\ 0.1 & 0.1 \\ -1.0 & -0.5 \\ 1.0 & 1.0 \\ 0.1 & 0.1 \\ 0.0 & 1.0 \end{bmatrix}$$

Problem 4: